Frozen spherical-cap field: radiated power vs frequency (BEM mesh m=2)
grey band: kR > 16.0, fewer than 6 elements per wavelength (unresolved)
dotted line: design point kR = 18.32 (100 Hz)

radiated power P (W)
(normal velocity frozen at the 100 Hz field)

10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}

sinc, spline-consistent [quadvel-rad]
sinc, centroid [quadvel-rad]
sinc on BEM mesh m=2 [quadvel-rad]
exact spectral two-sided (independent, novelty) [quadvel-rad]
BEM two-sided m=2, far-field power [quadvel-rad]
BEM far-field, UNRESOLVED (<6 el./wl) [quadvel-rad]
BEM surface power (not used) [quadvel-rad]
surface power off far field by >2 % (unreliable)
sinc, spline-consistent [uniform]
sinc, centroid [uniform]
sinc on BEM mesh m=2 [uniform]
exact spectral two-sided (independent, novelty) [uniform]
BEM two-sided m=2, far-field power [uniform]
BEM far-field, UNRESOLVED (<6 el./wl) [uniform]
BEM surface power (not used) [uniform]

power ratio sinc / two-sided (-)

10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}

consistent sinc / BEM far field [quadvel-rad]
centroid sinc / BEM far field [quadvel-rad]
spectral sinc / spectral two-sided [quadvel-rad]
consistent sinc / BEM far field [uniform]
centroid sinc / BEM far field [uniform]
spectral sinc / spectral two-sided [uniform]
breathing sphere: sinc / exact exterior (analytic)
spherical-exterior-benchmark.csv

black open circles: BEM point with < 6 elements per wavelength

share of two-sided power (-)
(symlog axis, linear for |y|<1)

10^{1}
10^{0}
0
-10^{0}
-10^{1}

P+ / P (outer, convex face) [quadvel-rad]
P- / P (inner, concave face) [quadvel-rad]
P_sinc,mesh / P (surface) [quadvel-rad]
P+ / P (outer, convex face) [uniform]
P- / P (inner, concave face) [uniform]
P_sinc,mesh / P (surface) [uniform]

10^{-1} 10^{0} 10^{1}

kR (R = 10 m radius of curvature of the cap; f = 343 kR / (20 pi) Hz)